

一、判断题

题号	1	2	3	4	5	6	7	8	9	10
答案	×	√	√	√	×	√	√	×	×	×

二、填空题

1. 5; 2. 32; 3. 2; 4. 1, -4;
5. $n!$

三、选择题

1. D; 2. B; 3. D; 4. A; 5. B.

四、计算题

$$\begin{aligned}
 \text{1. 解 } D &= \begin{vmatrix} 1 & 3 & -1 & 2 \\ 1 & -5 & 3 & -4 \\ 0 & 2 & 1 & -1 \\ -5 & 1 & 3 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -1 & 2 \\ 0 & -8 & 4 & -6 \\ 0 & 2 & 1 & -1 \\ 0 & 16 & -2 & 7 \end{vmatrix} \\
 &\quad \underline{\underline{\text{按第一列展开}}}. \begin{vmatrix} -8 & 4 & -6 \\ 2 & 1 & -1 \\ 16 & -2 & 7 \end{vmatrix} \\
 &= \begin{vmatrix} -16 & 4 & -2 \\ 0 & 1 & 0 \\ 20 & -2 & 5 \end{vmatrix} \\
 &= \begin{vmatrix} -16 & -2 \\ 20 & 5 \end{vmatrix} = -80 + 40 = -40. \quad \blacksquare
 \end{aligned}$$

$$\begin{aligned}
 \text{2. 解 法一: 由于 } (A|E) &= \begin{pmatrix} 1 & 3 & 0 & 1 & 0 & 0 \\ 2 & 6 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 & 1 & 0 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & -1 & 1 \\ 0 & 0 & 1 & -2 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & -5 & 3 & -3 \\ 0 & 1 & 0 & 2 & -1 & 1 \\ 0 & 0 & 1 & -2 & 1 & 0 \end{pmatrix}
 \end{aligned}$$

$$\text{所以 } A^{-1} = \begin{pmatrix} -5 & 3 & -3 \\ 2 & -1 & 1 \\ -2 & 1 & 0 \end{pmatrix}. \quad \blacksquare$$

法二：由于 $|A| = -1 \neq 0$,

$$A^* = \begin{pmatrix} 5 & -3 & 3 \\ -2 & 1 & -1 \\ 2 & -1 & 0 \end{pmatrix}$$

$$\text{所以 } A^{-1} = \frac{1}{|A|} A^* = \begin{pmatrix} -5 & 3 & -3 \\ 2 & -1 & 1 \\ -2 & 1 & 0 \end{pmatrix}. \quad \blacksquare$$

3.解 对矩阵 $(\alpha_1^T, \alpha_2^T, \alpha_3^T, \alpha_4^T)$ 作初等行变换, 得:

$$\begin{aligned} (\alpha_1^T, \alpha_2^T, \alpha_3^T, \alpha_4^T) &= \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & -2 \end{pmatrix} \\ &\rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 1 & -2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} = (\beta_1, \beta_2, \beta_3, \beta_4) \end{aligned}$$

$$\because r(\beta_1, \beta_2, \beta_3, \beta_4) = 2,$$

$\therefore$ 向量组 $\beta_1, \beta_2, \beta_3, \beta_4$ 的极大线性无关组含有两个向量.

显然, β_1, β_3 是向量组 $\beta_1, \beta_2, \beta_3, \beta_4$ 的一个极大线性无关组, 且

$$\beta_2 = 2\beta_1 + 0\beta_3, \quad \beta_4 = 2\beta_1 - 2\beta_3.$$

故 α_1, α_3 是 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 的一个极大线性无关组, 且

$$\alpha_2 = 2\alpha_1 + 0\alpha_3, \quad \alpha_4 = 2\alpha_1 - 2\alpha_3. \quad \blacksquare$$

$$4. \text{解} \text{ 由于 } (A|\mathbf{b}) = \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & 1 & -3 & 0 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 5 & 4 & 3 & 3 & -1 & 2 \end{array} \right) \rightarrow \left(\begin{array}{ccccc|c} 1 & 0 & -1 & -1 & -5 & -2a \\ 0 & 1 & 2 & 2 & 6 & 3a \\ 0 & 0 & 0 & 0 & 0 & 3-3a \\ 0 & 0 & 0 & 0 & 0 & 2-2a \end{array} \right)$$

所以当 $a=1$ 时, $r(A)=r(A|\mathbf{b})=2<4$, 原方程组有解, 且有无穷多解.

方法一: 此时, 原方程组的同解方程组为 $\begin{cases} x_1 - x_3 - x_4 - 5x_5 = -2, \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 3. \end{cases}$

令 $(x_3, x_4, x_5)^T = (0, 0, 0)^T$, 得特解为 $\boldsymbol{\eta}_0 = (-2, 3, 0, 0, 0)^T$.

导出方程组的同解方程组为 $\begin{cases} x_1 - x_3 - x_4 - 5x_5 = 0, \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 0. \end{cases}$ 显然导出方程组有无穷多解, 自由未知量个数为 3.

取 x_3, x_4, x_5 为自由未知量. 分别令 $(x_3, x_4, x_5)^T = (1, 0, 0)^T, (0, 1, 0)^T, (0, 0, 1)^T$, 得导出组的基础解系为:

$$\boldsymbol{\xi}_1 = (1, -2, 1, 0, 0)^T, \boldsymbol{\xi}_2 = (1, -2, 0, 1, 0)^T, \boldsymbol{\xi}_3 = (5, -6, 0, 0, 1)^T.$$

故原方程组的通解为:

$$\boldsymbol{\eta} = \boldsymbol{\eta}_0 + k_1 \boldsymbol{\xi}_1 + k_2 \boldsymbol{\xi}_2 + k_3 \boldsymbol{\xi}_3$$

$$= \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + k_3 \begin{pmatrix} 5 \\ -6 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad k_1, k_2, k_3 \text{ 为任意常数.} \quad \blacksquare$$

方法二: 此时, 原方程组的同解方程组为 $\begin{cases} x_1 - x_3 - x_4 - 5x_5 = -2, \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 3. \end{cases}$

自由未知量个数为 3, 取 x_3, x_4, x_5 为自由未知量, 令 $x_3 = k_1, x_4 = k_2, x_5 = k_3$, 得原方程组的通解为:

$$\begin{cases} x_1 = -2 + k_1 + k_2 + 5k_3 \\ x_2 = 3 - 2k_1 - 2k_2 - 6k_3 \\ x_3 = k_1 \\ x_4 = k_2 \\ x_5 = k_3 \end{cases} \quad (k_1, k_2, k_3 \text{ 为任意常数}). \quad \blacksquare$$

5. 解 由

$$|\lambda E - A| = \begin{vmatrix} \lambda - 3 & -1 & 0 \\ -1 & \lambda - 3 & 0 \\ 0 & 0 & \lambda - 2 \end{vmatrix} = (\lambda - 2)^2 (\lambda - 4) = 0$$

得特征值为 $\lambda_1 = \lambda_2 = 2, \lambda_3 = 4$.

当 $\lambda_1 = \lambda_2 = 2$ 时, 解 $(2E - A)\mathbf{x} = \mathbf{0}$.

$$\text{因 } 2E - A = \begin{pmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得基础解系 } \xi_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

当 $\lambda_3 = 4$ 时, 解 $(4E - A)\mathbf{x} = \mathbf{0}$.

$$\text{因 } 4E - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得基础解系 } \xi_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

由于 A 有三个线性无关的特征向量, 故能相似对角化;

$$\text{相似变换矩阵 } P = (\xi_1, \xi_2, \xi_3) = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \Lambda = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} \quad \blacksquare$$

五、证明题 (

证 由 $A^3 - A^2 + 2A - E = O$ 可得:

$$A^3 - A^2 + 2A = E,$$

$$\text{即 } A^2(A - E) + 2(A - E) = -E$$

$$\text{即 } (A^2 + 2E)(A - E) = -E$$

$$\text{即 } (-A^2 - 2E)(A - E) = E$$

从而可知 $A - E$ 可逆, 且 $(A - E)^{-1} = -A^2 - 2E$. $\blacksquare$